Exercise 14

Classes and Encapsulation: Bank Account Management

Introduction:

As a program's code base grows in size, so does its complexity. When the program is considered as a collection of autonomous entities, each responsible for its own data and behavior, that complexity can be managed effectively. In C++, a class is the mechanism by which encapsulation is achieved: data members are declared private, and access to them is granted only through public member functions.

A class is declared using the class keyword. Private members are accessible only within the class itself, while public members define the interface available to the rest of the program. The basic structure follows the syntax below:

```
class ClassName {
  private:
    datatype memberName;
  public:
    returnType memberFunction(paramType paramName);
};
```

Member functions defined outside the class body use the scope resolution operator to associate them with their class. An object of the class is instantiated in function main() and its public member functions are called through that object.

```
returnType ClassName::memberFunction(paramType paramName) { ... }
```

Learning Outcomes:

- Implement a class with private data members and public member functions that model a bank account.
- Apply the class interface to perform deposits, withdrawals, and balance inquiries through an instantiated object.

Problem Description:

```
/* Constructor that initializes the account with the owner's name ownerName
   and a starting balance of initialBalance. Sets balance to 0 if initialBalance
   is negative. */
BankAccount(string ownerName, double initialBalance);

/* Adds amount to the current balance if amount is greater than zero.
   Prints a confirmation message indicating the new balance. */
void deposit(double amount);
```

```
/* Deducts amount from the current balance if amount is greater than zero
   and does not exceed the available balance. Prints a message indicating
   success or insufficient funds. */
void withdraw(double amount);

/* Returns the current balance of the account. */
double getBalance() const;

/* Prints the account owner's name and current balance to standard output
   in a formatted summary. */
void printSummary() const;
```

Call all the functions created in function main().